

ARK-458 — Results

Cloud IAM Role Grant · Exact-Action Binding

Verdict: PASS

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Substrate: Classical software (no quantum hardware, no cryptographic security claim)

Guards: Dual independent — V1 (JavaScript), V2 (Python)

Headline

- `g_allow_exact` = **1.0000** ($C1 \geq 0.95$ ✓)
- `g_deny_mutation_min` = **1.0000** ($C2 \geq 0.95$ ✓)
- `g_margin` = **0.9500** ($C3 \geq 0.90$ ✓)
- Dual-guard concordance: **800/800 = 100.00%**
- Every one of the 700 mutated IAM actions across arms 2–8 was **DENIED**; every exact match in arm 1 was **ALLOWED**.

Per-arm readout (100 scenarios each)

Arm	Condition	Expected	V2 ALLOW	V2 DENY	V1↔V2 concordance
1	BASELINE-exact-action-match	ALLOW	100	0	100%
2	ROLE-ESCALATION	DENY	0	100	100%
3	PRINCIPAL-SWAP	DENY	0	100	100%
4	ACCOUNT-SWAP	DENY	0	100	100%
5	PERMISSION-EXPANSION	DENY	0	100	100%
6	CONDITION-STRIP	DENY	0	100	100%
7	MULTI-DIM-TAMPER	DENY	0	100	100%
8	CONFUSABLE-NEAR-MATCH	DENY	0	100	100%

Kill-gate (negative control)

A deliberately lenient comparator (case/whitespace-folding, homoglyph-mapping, ignoring the effective-policy hash, and accepting any “at-least-as-privileged” role) was run against the same 700 attack scenarios. It **wrongly ALLOWED 281/700** mutated actions:

Arm	Bad-guard wrong-allows
2 ROLE-ESCALATION	100
5 PERMISSION-EXPANSION	100
7 MULTI-DIM-TAMPER	5
8 CONFUSABLE-NEAR-MATCH	76

This confirms the testbed is **falsifiable** — the arms are genuine attacks that a naive guard fails, and the exact-action-binding guard’s clean sweep is meaningful rather than trivial. (Full detail in `results/killgate_results.json`.)

Effectiveness gate

All 800 scenarios passed the effectiveness oracle: every attack arm genuinely mutated the action, arm 2 was a strict privilege escalation in every case, arm 8 was confusable-under-normalization in every case, and arm 1 was a true exact match.

What this shows (bounded)

Within this scenario model, an ExecutionProof exact-action-binding guard authorizes an IAM role grant **only** when the execution action is byte-identical to the approved action across all five binding dimensions, and **fails closed** on every mutation — including the two failure modes a lenient comparator most often gets wrong: **privilege escalation** and **confusable near-matches**. Two independent implementations agreed on all 800 decisions.

Honest bounds / what this does NOT show

- This is a classical test of **authorization control logic**, not a test of AWS IAM itself, not a cryptographic security proof, and not a production-readiness or RF-100 conformance certification.
- `permission_set` is a plain SHA-256 identity token for “the exact policy body,” not a security primitive.
- Results are bounded to this scenario model, these eight arms, and this seed family.
- This experiment provides dated **working-example** evidence for disclosed mechanisms; it does **not** legally validate any patent claim and adds no new matter to any filed application.

Reproduce

```
cd ark-458
node --version          # dual-guard V1 requires Node
python3 run_killgate.py  # negative control
python3 run_arms.py      # 8 arms × 100 scenarios, dual guards
./compute_hashes.sh      # verify locked files match MANIFEST.txt
```

Artifacts: `results/arm_1_results.json` ... `results/arm_8_results.json` , `results/over-all_results.json` , `results/killgate_results.json` , and per-arm `results/arm_N_scenarios.json` .

To God be the glory. Proof Before Power. Verification Before Execution.